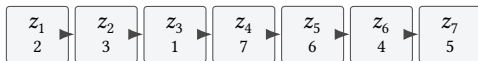


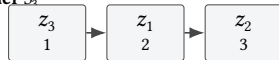
(a) Input curve order



valid sequence: (2, 3, 1, 7, 6, 4, 5)

(b) Initialization

partial order S_2



upper family
 $P_2 = \{z_1, z_2\}$

lower family
 $P_3 = \{z_2, z_3\}$

(c) Representative iteration $i = 4$

$S_3 = (z_3, z_1, z_2)$
values (1, 2, 3)

new point $z_4 = 7$

Step 3(a)
add $P_4 = \{z_3, z_4\}$

Step 3(b)
transfer P_2 to P_4

Step 3(c)
insert after anchor z_2

$S_4 = (z_3, z_1, z_2, z_4)$ with values (1, 2, 3, 7)

(d) Recovered axis order



output from maintained partial order: (1, 2, 3, 4, 5, 6, 7)